

Assignment 12

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May 15, 2026

1 Page 171, Problem 54

Given that $X \sim N(\mu_X, \sigma_X^2)$, $Y \sim N(\mu_Y, \sigma_Y^2)$, $Z \sim N(\mu_Z, \sigma_Z^2)$, where they are uncorrelated, for

$$\begin{aligned}U &= Z + X \\V &= Z + Y\end{aligned}$$

we have

$$\begin{aligned}\text{Cov}(U, V) &= \text{Cov}(Z + X, Z + Y) \\&= \text{Cov}(Z, Z) + \text{Cov}(Z, Y) + \text{Cov}(X, Z) + \text{Cov}(X, Y) \\&= \sigma_Z^2\end{aligned}$$

and

$$\begin{aligned}\rho_{UV} &= \frac{\text{Cov}(U, V)}{\sqrt{\text{Var}(U)\text{Var}(V)}} \\&= \frac{\sigma_Z^2}{\sqrt{(\sigma_X^2 + \sigma_Z^2)(\sigma_Y^2 + \sigma_Z^2)}}\end{aligned}$$

2 Page 171, Problem 60

Given that the density function have the symmetric property $f(y) = f(-y)$ then $E(Y) = 0$ and $X = SY$, where S is a random variable independent variable taking on the value $+1$ and -1 with probability $\frac{1}{2}$, we have

$$\begin{aligned}\text{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\&= E(SY^2) - E(S)E(Y)E(Y) \\&= E(S)E(Y^2) - E(S)E(Y)E(Y) \\&= 0\end{aligned}$$

We show that X and Y are not independent. Since $X = SY$,

$$\begin{aligned}P(X = Y) &= P(S = 1) + P(S = -1, Y = 0) \\&= \frac{1}{2} + 0 = \frac{1}{2},\end{aligned}$$

because Y has a density and thus $P(Y = 0) = 0$. If X and Y were independent and each had a density, then $P(X = Y) = 0$, a contradiction. Hence X and Y are not independent.

3 Extra Question

3.1 1)

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} \frac{x}{2} e^{-|x|} dx \\ &= \int_0^{\infty} \frac{x}{2} e^{-x} dx + \int_{-\infty}^0 \frac{x}{2} e^x dx \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 \\ &= \int_{-\infty}^{\infty} \frac{x^2}{2} e^{-|x|} dx \\ &= \int_0^{\infty} \frac{x^2}{2} e^{-x} dx + \int_{-\infty}^0 \frac{x^2}{2} e^x dx \\ &= 2 \end{aligned}$$

3.2 2)

No, X and $|X|$ are not independent. If we know the value of $|X|$, the value of X is completely restricted to $\pm|X|$. For instance,

$$P(X \geq 1 \mid |X| < 1) = 0$$

but $P(X \geq 1) > 0$. Since the conditional probability is not equal to the marginal probability, they are dependent.

3.3 3)

They are uncorrelated. To check for correlation, we compute the covariance:

$$\text{Cov}(X, |X|) = E(X|X|) - E(X)E(|X|)$$

We already computed that $E(X) = 0$. Furthermore,

$$E(X|X|) = \int_{-\infty}^{\infty} x|x| \frac{1}{2} e^{-|x|} dx$$

Notice that $g(x) = x|x|$ is an odd function, and the density $f(x) = \frac{1}{2}e^{-|x|}$ is an even function. Their product is an odd function. Integrating an odd function over the symmetric interval $(-\infty, \infty)$ gives 0. Hence, $E(X|X|) = 0$.

Therefore,

$$\text{Cov}(X, |X|) = 0 - 0 \cdot E(|X|) = 0$$

Since the covariance is zero, X and $|X|$ are uncorrelated (not correlated).

4 Supplementary Question 1.1

$$f_X(x) = \int_0^2 \frac{x+y}{8} dy = \frac{x}{4} + \frac{1}{4}$$

$$f_Y(y) = \int_0^2 \frac{x+y}{8} dx = \frac{y}{4} + \frac{1}{4}$$

$$E(X) = \int_0^2 x \left(\frac{x}{4} + \frac{1}{4} \right) dx = \frac{7}{6}$$

$$E(Y) = \int_0^2 y \left(\frac{y}{4} + \frac{1}{4} \right) dy = \frac{7}{6}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \int_0^2 \int_0^2 xy \frac{x+y}{8} dx dy - \frac{7}{6} \cdot \frac{7}{6} = \frac{4}{3} - \frac{49}{36} = -\frac{1}{36}$$

$$\begin{aligned} \rho_{XY} &= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \\ &= \frac{-\frac{1}{36}}{\sqrt{\left(\int_0^2 x^2 \left(\frac{x}{4} + \frac{1}{4} \right) dx - \left(\frac{7}{6} \right)^2 \right) \left(\int_0^2 y^2 \left(\frac{y}{4} + \frac{1}{4} \right) dy - \left(\frac{7}{6} \right)^2 \right)}} \\ &= \frac{-\frac{1}{36}}{\sqrt{\left(\frac{5}{3} - \frac{49}{36} \right)^2}} \\ &= -\frac{1}{11} \end{aligned}$$

$$\begin{aligned} D(X+Y) &= D(X) + D(Y) + 2\text{Cov}(X, Y) \\ &= \frac{11}{36} + \frac{11}{36} + 2 \cdot \left(-\frac{1}{36} \right) \\ &= \frac{5}{9} \end{aligned}$$

5 Supplementary Question 1.2

Given that $X, Y \sim N(\mu, \sigma^2)$ and X, Y are independent. Let

$$Z = \alpha X + \beta Y, W = \alpha X - \beta Y$$

then

$$\begin{aligned} \text{Cov}(Z, W) &= E(ZW) - E(Z)E(W) \\ &= E((\alpha X + \beta Y)(\alpha X - \beta Y)) - E(\alpha X + \beta Y)E(\alpha X - \beta Y) \\ &= E(\alpha^2 X^2 - \beta^2 Y^2) - E(\alpha X + \beta Y)E(\alpha X - \beta Y) \\ &= \alpha^2 E(X^2) - \beta^2 E(Y^2) - (\alpha E(X) + \beta E(Y))(\alpha E(X) - \beta E(Y)) \\ &= \alpha^2(\sigma^2 + \mu^2) - \beta^2(\sigma^2 + \mu^2) - (\alpha\mu + \beta\mu)(\alpha\mu - \beta\mu) \\ &= (\alpha^2 - \beta^2)(\sigma^2 + \mu^2) - (\alpha^2 - \beta^2)\mu^2 \\ &= (\alpha^2 - \beta^2)\sigma^2 \end{aligned}$$

$$\begin{aligned} \rho_{ZW} &= \frac{\text{Cov}(Z, W)}{\sqrt{\text{Var}(Z)\text{Var}(W)}} \\ &= \frac{(\alpha^2 - \beta^2)\sigma^2}{\sqrt{(\alpha^2\sigma^2 + \beta^2\sigma^2)(\alpha^2\sigma^2 + \beta^2\sigma^2)}} \\ &= \frac{(\alpha^2 - \beta^2)\sigma^2}{\sqrt{(\alpha^2 + \beta^2)^2\sigma^4}} \\ &= \frac{\alpha^2 - \beta^2}{\alpha^2 + \beta^2} \end{aligned}$$

6 Question 1

Let

$$Z = X - Y$$

Thus $E(Z) = 2 - 2 = 0$, and

$$\begin{aligned}\text{Var}(Z) &= \text{Var}(X) + \text{Var}(Y) - 2\rho\sqrt{\text{Var}(X)\text{Var}(Y)} \\ &= 1 + 4 - 2 \cdot \frac{1}{2} \cdot \sqrt{1 \cdot 4} \\ &= 3\end{aligned}$$

Thus

$$P(|X - Y| \geq 6) = P(|Z - E(Z)| \geq 6) \leq \frac{\text{Var}(Z)}{6^2} = \frac{1}{12}$$

7 Question 2

Let $X_1, X_2, \dots, X_n$ be the number of dice rolled, respectively. The arithmetic mean of the number of dice rolled is $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$. We have $E(X_i) = 3.5$. By the weak law of large numbers, the mean $\bar{X}$ converges in probability to $E(X_i) = 3.5$.

8 Question 3

Let the random variable $X \sim EXP(100)$, picking 16 samples randomly from the components. Thus,

$$\begin{aligned}P\left(\sum_{i=1}^{16}(X_i) > 1920\right) &= P\left(\left(\frac{\sum_{i=1}^{16}(X_i) - 16 \cdot E(X)}{\sqrt{16 \cdot \text{Var}(X)}} > \frac{1920 - 16 \cdot E(X)}{\sqrt{16 \cdot \text{Var}(X)}}\right)\right) \\ &= P\left(\frac{\sum_{i=1}^{16}(X_i) - 16 \cdot E(X)}{\sqrt{16 \cdot \text{Var}(X)}} > \frac{1920 - 1600}{400}\right) \\ &= P\left(\frac{\sum_{i=1}^{16}(X_i) - 16 \cdot E(X)}{\sqrt{16 \cdot \text{Var}(X)}} > 0.8\right) \\ &= 1 - P\left(\frac{\sum_{i=1}^{16}(X_i) - 16 \cdot E(X)}{\sqrt{16 \cdot \text{Var}(X)}} \leq 0.8\right) \\ &= 1 - \Phi(0.8) \\ &\approx 0.2119\end{aligned}$$

9 Question 4

Let S be the number of death in a year, $E(S) = 10000 \times 0.017 = 170$. $\text{Var}(S) = 10000 \times 0.017 \times 0.983 = 167.11$.

Thus,

$$\begin{aligned}P(S > 200) &= P\left(\frac{S - E(S)}{\sqrt{\text{Var}(S)}} > \frac{200 - 170}{\sqrt{167.11}}\right) \\ &= P\left(\frac{S - E(S)}{\sqrt{\text{Var}(S)}} > 2.32\right) \\ &= 1 - P\left(\frac{S - E(S)}{\sqrt{\text{Var}(S)}} \leq 2.32\right) \\ &= 1 - \Phi(2.32) \\ &\approx 0.0102\end{aligned}$$

10 Supplementary Question 2.1

For X_k , $P(X_k = \sqrt{\ln k}) = \frac{1}{2}$, and $P(X_k = -\sqrt{\ln k}) = \frac{1}{2}$. Thus, $E(X_k) = 0$ and $\text{Var}(X_k) = \ln k$. We prove that it follows the law of large numbers by checking the sufficient condition.

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \text{Var}(X_k) &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n \ln k \\ &\leq \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot n \ln n \\ &= \lim_{n \rightarrow \infty} \frac{\ln n}{n} \\ &= 0\end{aligned}$$

Thus, by the sufficient condition of the weak law of large numbers, we have $\bar{X} = \frac{1}{n} \sum_{k=1}^n X_k$ converges in probability to $E(X_k) = 0$.

11 Supplementary Question 2.2

For $E(X_n) = p_n$, $\text{Var}(X_n) = p_n(1 - p_n)$. We check the sufficient condition of the weak law of large numbers:

$$\begin{aligned}\lim_{n \rightarrow \infty} \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n p_i(1 - p_i) \\ &\leq \lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \frac{1}{4} \\ &= \lim_{n \rightarrow \infty} \frac{1}{4n} \\ &= 0\end{aligned}$$

Thus, by the sufficient condition of the weak law of large numbers, we have $\bar{X} = \frac{1}{n} \sum_{k=1}^n X_k$ converges in probability to $E(X_k) = 0$.